

Means-Tested Pensions and Monotone Policies: Saving Rises, Consumption Need Not

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Abstract

A pension that is withdrawn as the recipient's assets rise is widely said to make the household's policy function non-monotone, and the reason usually given is that the means test makes the value function non-concave. We show that the conclusion is right, the reason is wrong, and that which policy misbehaves depends on the state space. With cash on hand as the state, optimal saving rises with the state for *every* continuation value, concave or not: only strict concavity of period utility is used, and the continuation cancels. What a non-concave continuation produces is a jump, and the jump falls on consumption, which drops discretely at the wealth where the household gives the pension up. With assets and the earnings shock as the state, next period's assets rise with this period's at each shock, again for every continuation; the repository's earlier version of that statement assumed concavity, which is exactly what a means test destroys. Monotone saving fails only when the test is assessed on assets currently held and withdraws the pension faster than the gross return, because then cash on hand itself falls as assets rise. A calibrated retiree confirms each case, and Australia's asset test sits on the benign side of the threshold with a non-monotone consumption function. Every result is checked by the Lean 4 proof assistant.

JEL: D15, H55, C63. *Keywords*: means testing, monotone comparative statics, consumption, borrowing constraints, formal verification, Lean.

1 Introduction

A pension that is withdrawn as the recipient's assets rise gives the household a reason to hold fewer assets, and it is often said that this makes the household's policy function non-monotone. The claim is usually supported by an appeal to the value function: the means test puts a kink, or a cliff, into the reward from saving, the value function is no longer concave, and monotone comparative statics are said to fail with it.

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. Comments welcome.

Three quotations fix what is actually claimed. Kudrna and Woodland (2011), simulating the Australian age pension in a general equilibrium life-cycle model, treat the geometry as settled: “the means test for the age pension causes the budget set to be non-convex, as is well known.” They pay for it in their solution method, whose solver “is not guaranteed to yield a globally optimal solution for the inter-temporal expected utility maximization problem when the budget constraint is non-convex”; they check the solution of every household over the age of sixty by hand to confirm that the reported optimum is global. Garstenauer and Siassi (2024), modelling the United States tax and transfer system for married couples with children, reach the same conclusion about method: “the presence of kinks and non-differentiabilities in the budget constraint generated by the tax-transfer system renders Euler equation methods inapplicable.” They solve by discrete-state value function iteration, and because “of the presence of these kinks and non-convexities, it is crucial for us to capture as accurately as possible how labor supply choices respond to taxes and transfers” they put mothers’ annual hours on a grid whose adjacent nodes are twenty-five hours apart.

The third is closest to our question, and it cuts the other way. Wellschmied (2021) studies an asset means test in a life-cycle model and characterises the policy analytically. His Theorem 1 states that the saving policy “is increasing”, and his proof that the value function stays sub-differentiable uses that the policy “is weakly increasing”. What he shows fails is everything else. “Despite the policy $k_h(\cdot, X)$ being monotone, the marginal propensity to consume out of additional assets may be one for households with low income and assets”; “the policy function makes a jump”; and “choices are distorted, despite satisfying first order conditions because of the kinks in the expected value function”. A footnote records how fast this compounds, since there is “a rapid increase in the number of non-differentiabilities in the value function”. The cost lands in the numerical appendix, where “first order conditions are not necessarily unique”, so off-grid choice is abandoned over the part of the state space the test touches and almost nine thousand asset nodes are carried instead. His argument works through the differential structure of the value function, following Clausen and Strub (2020) on how kinks propagate under maximisation; the argument below uses no calculus at all.

The conclusion is right and the reason is wrong, and Wellschmied (2021) is already halfway to saying so. Monotonicity of the saving policy in cash on hand survives *any* continuation value whatever — concave, kinked, discontinuous, or produced by a means test of any shape — because the only structure the comparison needs is strict concavity of period utility, and the continuation cancels between the two optimality inequalities. This is Topkis’s argument (Topkis, 1978; Milgrom and Shannon, 1994) applied to the household’s problem, and it does not care what the continuation looks like.

What a non-concave continuation does produce is a jump. At the wealth where the household stops protecting its pension and starts saving past the test, optimal saving jumps up, and because it jumps by more than wealth has risen, consumption jumps *down*. So the non-monotonicity is real, it is in consumption, and it coexists with a perfectly monotone saving policy. We give an exact two-period example, with the optimum at each of two wealth levels computed and verified.

Which policy misbehaves also depends on which state space one works in, and the two are not equivalent under a means test. In the household's own state space — assets a and the earnings shock z — next period's assets rise with this period's at each z , again for every continuation. The Lean development that this paper draws on already had that statement, but with concavity of the continuation as a hypothesis; a means test is exactly what removes that hypothesis's premise, so we prove it without. Monotone saving in assets then fails in one case only: when the test is assessed on the assets currently held and withdraws the pension faster than the gross return, cash on hand itself falls as assets rise, and the policy follows it down. The threshold is sharp, the taper rate against the gross return, and Australia's asset test sits on the benign side of it.

Section 2 sets out the retired household and the three pension schedules. Section 3 gives the two results in the cash-on-hand state space: saving monotone always, consumption not. Section 4 gives the result in the (a, z) state space and the exception. Section 5 calibrates a retiree and checks each case. Every result stated with a formal name has been checked by the Lean 4 proof assistant (de Moura and Ullrich, 2021) against Mathlib (The mathlib Community, 2020); the development is at <https://github.com/LeanEconomics/LeanEconomics>, files `LeanEconomics/OLG/MeansTest.lean` and `Retirement.lean`.

2 The retired household

A household lives J periods, works until age J_r and is retired after it. Period utility is CRRA with coefficient μ and discount factor β , the gross return is $R = 1 + r$, and assets may not fall below a borrowing limit, taken to be zero. In retirement labour earnings are replaced by a pension.

A lump-sum pension. If the pension is a constant p , the retired household faces the same problem as the working one with earnings replaced by p , so the retired phase is an income-fluctuation problem in its own right (`withPension`) and the whole of the standard theory applies to it. Two facts are worth recording. A retiree's human wealth does not depend on the earnings shock, because the pension does not, and it never exceeds the perpetuity value of the pension, $H_k \leq p/r$ (`stageHumanWealth_withPension_const`, `stageHumanWealth_withPension_1e`). And cash on hand is strictly increasing in assets (`withPension_resources_strictMono`). Both fail once the pension is means-tested.

Three schedules. Write the pension as a function $p(\cdot)$ of assets and cash on hand as

$$m(a) = Ra + p(a)$$

(cash). We consider three schedules (`lumpSum`, `tapered`, `cliff`):

- a *lump sum*, $p(a) = p$;

- a *taper*, $p(a) = \max\{0, p - \tau \max\{0, a - \bar{a}\}\}$, which withdraws the pension at rate τ per unit of assets above a free area $\bar{a}$ and never pays a negative amount;
- a *cliff*, $p(a) = p$ for $a \leq \bar{a}$ and 0 above, the limit of a taper as $\tau \rightarrow \infty$.

Where the test is assessed. It matters whether the pension paid this period depends on the assets held this period or on the assets carried into the next. A test on assets *carried forward* enters only the continuation value, leaving today's budget alone. A test on assets *currently held* enters today's cash on hand directly. Real asset tests are of the second kind, assessed each period on current holdings, and the two have different consequences, as Sections 3 and 4 show.

One period of the problem. Throughout we write the household's choice as

$$\max_{x \in [0, m]} u(m - x) + W(x), \quad (1)$$

with m cash on hand, x next period's assets and W the continuation value, and call x *optimal* at m when it attains the maximum (`IsOptimalSaving`). The feasible set excludes $x = m$, so consumption is positive. Nothing below assumes anything about W beyond what is stated: in particular it is never assumed concave, continuous, or generated by any particular pension schedule.

3 Cash on hand as the state

3.1 Saving is monotone, whatever the continuation

Theorem 3.1 (Monotone saving in cash on hand). *Let period utility be strictly concave on the positive half-line and let W be any continuation value. If $m_1 < m_2$ and x_i is optimal at m_i in (1), then $x_1 \leq x_2$.*

Formal statement: `OLG.isOptimalSaving_mono`.

Proof sketch. Suppose $x_2 < x_1$. Both choices are feasible at both states, so adding the two optimality inequalities,

$$u(m_1 - x_1) + u(m_2 - x_2) \geq u(m_1 - x_2) + u(m_2 - x_1),$$

the continuation terms cancelling. Write $p = m_1 - x_1$ and $q = m_2 - x_2$. Since $x_2 < x_1$ and $m_1 < m_2$ we have $p < m_1 - x_2 < q$, and $m_2 - x_1 = p + q - (m_1 - x_2)$ also lies strictly between p and q . The two right-hand arguments are therefore strict convex combinations of p and q with complementary weights, so strict concavity gives $u(m_1 - x_2) + u(m_2 - x_1) > u(p) + u(q)$, contradicting the display. $\square$

The proof uses nothing about W : it enters both optimality inequalities once and cancels. So a value function made non-concave by a means test — or by anything else — cannot by itself make saving fall with cash on hand. Note also what the hypotheses do *not* include:

no differentiability, no interiority, no uniqueness of the optimum. The conclusion holds for arbitrary optimal selections, which matters here, because at the switch point the optimum is genuinely multi-valued.

3.2 Consumption is not monotone

What a non-concave continuation does produce is a jump in saving, and the jump is larger than the increase in wealth that caused it.

Theorem 3.2 (Consumption falls when wealth rises). *There are a continuation value arising from a means-tested pension and two levels of cash on hand, the second larger, at which optimal consumption is strictly smaller.*

Formal statement: *OLG.exists_consumption_decrease.*

The witness is exact. Take logarithmic utility, a unit gross return, and a pension of one withdrawn entirely above assets of one, so that next period's cash on hand is

$$N(x) = x + \mathbf{1}\{x \leq 1\},$$

which jumps down by one as x crosses the threshold, and let $W = \log N$. Then:

Cash on hand m	optimal saving x	consumption $m - x$	next period's cash $N(x)$
6	1	5	2
8	4	4	4

At $m = 6$ the household keeps the pension, saving the most it can without losing it; at $m = 8$ the pension is no longer worth protecting, so the household gives it up and saves the proceeds. Saving rises from 1 to 4, as Theorem 3.1 says it must, and since it rises by more than wealth does, consumption falls from 5 to 4.

Both optima are verified rather than asserted. Because utility is logarithmic, comparing $\log(m - x) + \log N(x)$ across x is comparing the product $(m - x)N(x)$, which is a quadratic on each side of the threshold. At $m = 6$ the product is $-x^2 + 5x + 6$ below the threshold, increasing there and equal to 10 at $x = 1$, and $(6 - x)x \leq 9$ above it. At $m = 8$ it is $-x^2 + 7x + 8 \leq 14$ below and $(8 - x)x \leq 16$ above, attained at $x = 4$. The comparisons are checked by the kernel (`isOptimalSaving_six`, `isOptimalSaving_eight`).

3.3 The two policies come apart

The contrast is not an artefact of proof technique. Consumption is $m - x$ with both terms rising in m , and when the continuation is non-concave the second can rise faster. Monotone consumption therefore genuinely requires concavity of the continuation, while monotone saving does not; the Lean development records the identity $c = m - x$ for exactly this purpose (`consumptionFnOf_eq_sub`) and derives consumption monotonicity only under concave slices. So in the cash-on-hand state space the means test leaves the saving policy alone and breaks the consumption policy, and the informal argument quoted in the introduction gets the object wrong as well as the reason.

4 Assets and the earnings shock as the state

The household's own state space is the pair (a, z) , assets and the earnings shock, with z following a finite Markov chain and cash on hand $m(a, z) = y(z) + Ra$ in the absence of a means test on current holdings. The question is then about $g_v(a, z)$, next period's assets.

4.1 Next period's assets rise with this period's

Theorem 4.1 (Monotone saving in assets). *Let $a < a'$ both lie in the asset region. Then for every continuation v and every z , $g_v(a, z) \leq g_v(a', z)$.*

Formal statement: *IncomeFluctuation.policyOf_mono_of_lt.*

This is Theorem 3.1 read in the household's state space: cash on hand rises strictly with assets, so optimal saving does. The development already contained this statement (`policyOf_mono`) but with concave slices of the continuation as a hypothesis, and a means test is precisely what destroys concavity, so the hypothesis had to go. The Topkis argument removes it, at the cost of requiring the two asset levels to be distinct, which is harmless. As in Section 3, the continuation cancels and only strict concavity of period utility survives.

Consumption does not inherit it, for the same reason as before: $c = m(a, z) - g_v(a, z)$ with both terms rising in a , and the witness of Theorem 3.2 transfers directly, since at fixed z cash on hand is an increasing affine function of assets.

4.2 When saving in assets does fail

Theorem 4.1 has one hypothesis that a means test can remove: that cash on hand rises with assets. Where the test is assessed on assets carried forward, today's budget is untouched and the theorem applies verbatim. Where it is assessed on assets currently held, the schedule enters cash on hand directly, and whether it still rises is a question about the taper rate.

Theorem 4.2 (A gentle taper is harmless; a cliff is not). *If $0 \leq \tau < R$ then $a \mapsto Ra + p(a)$ is strictly increasing under the taper (`cash_strictMono_tapered`), and hence, by Theorem 3.1, saving is monotone in assets (`isOptimalSaving_mono_assets`). Under a cliff it is not: if $a \leq \bar{a} < b$ and $R(b - a) < p$ then cash on hand is strictly smaller at b (`cash_lt_cash_of_cliff`), and there are two asset levels, the second larger, at which the household saves strictly less.*

Formal statement: *OLG.exists_saving_decrease_in_assets.*

The threshold is sharp and it is the taper rate against the gross return. A pension withdrawn more slowly than assets earn leaves cash on hand rising, and then no amount of damage to the value function can make saving fall; a pension withdrawn faster makes a household with more assets poorer in the only sense that matters for today's decision, and saving falls with it. The proof of the first half is that the positive part is one-Lipschitz, so the pension falls by at most τ per unit of assets while gross assets rise by R (`max_zero_sub_1e`).

The witness for the second half is again exact, and deliberately built on a *concave* continuation so that nothing is hidden in it. With a unit gross return, a pension of one withdrawn above assets of one, and continuation $\log(x + 1)$:

Assets a	cash on hand $Ra + p(a)$	optimal saving	consumption
1	2	1/2	3/2
3/2	3/2	1/4	5/4

The household with more assets has less to spend, saves less and consumes less. The non-monotonicity here lives in the budget constraint, not in the shape of the value function, and that is the whole difference between this case and Section 3.

5 A calibrated retiree

The theorems predict exactly which of the two channels operates for which schedule, and a calibration confirms it case by case. We solve a deterministic retiree of 25 years by backward induction with a global search over a grid of 6001 asset levels — the value function is not concave, so a first-order condition cannot be used — at $r = 4\%$, $\beta = 0.96$, relative risk aversion 3, a full pension of 0.40 of mean earnings and a free area of 3 mean-earnings units (`numerics/meanstest.m`). The test is assessed on assets currently held, so all three channels are live. The taper of 0.078 per unit of assets is Australia’s asset test, three dollars a fortnight per thousand dollars, the schedule studied by Kudrna and Woodland (2011).

Pension schedule	Cash on hand	Saving	Consumption
Lump sum	increasing	increasing	increasing
Taper 0.078 (Australia)	increasing	increasing	falls by 0.025–0.046
Taper 0.50	increasing	increasing	falls by 0.13–0.24
Taper 1.20 (above R)	falls	falls by 0.004	falls by 0.55–0.70
Cliff	falls by 0.40	falls by 0.32–0.37	falls by 0.55–0.70

Every entry is what the theorems require. Saving falls with assets in exactly the two rows where cash on hand does, which are exactly the two rows whose taper exceeds the gross return, as Theorem 4.2 says; in the other three it rises, as Theorem 4.1 says it must. Consumption falls in every row with a test of any kind, as Theorem 3.2 says it can, and only in the lump-sum row is every policy monotone.

The consumption drop sits exactly at the free area and grows with age, from 0.025 of mean earnings at the start of retirement to 0.046 twenty years in under the Australian taper. The mechanism is the one the witness of Section 3 isolates: just above the free area the effective return to saving is only $R - \tau$ until the pension is exhausted, so the household jumps its saving towards the far side of the taper range and cuts consumption to pay for it. Under the Australian schedule the borrowing constraint binds at about eight per cent of the state space at every age.

The effect is not confined to retirees. A worker in the last year before retirement, facing the Australian taper for the whole of retirement, has a monotone saving policy and a consumption policy that falls by 0.025 of mean earnings at the same threshold: the non-concavity is inherited through the continuation value, and it is again consumption that carries it. That

is Theorem 3.1 at work — the continuation is as badly behaved as one likes, and saving is monotone anyway.

6 Conclusion

Three statements, in the order they should be made.

Optimal saving rises with cash on hand for every continuation value, so a value function made non-concave by a means test cannot by itself reverse a policy. The common argument from non-concavity to non-monotonicity is therefore not valid as stated, and it is not a matter of sharper hypotheses: the continuation cancels.

Consumption is a different object and it is genuinely non-monotone. At the wealth where the household stops protecting its pension, saving jumps up by more than wealth has risen and consumption drops. We give the drop exactly in a two-period example and measure it in a calibrated retiree, where Australia’s asset test produces a fall of two and a half to five per cent of mean earnings at the free area, at every age of retirement and in the last year of work as well.

Next period’s assets rise with this period’s, at each earnings shock, again for every continuation. This fails in one case only: when the test is assessed on the assets currently held and withdraws the pension faster than assets earn, cash on hand itself falls and the policy follows. The threshold is the taper rate against the gross return, and it is sharp.

For applied work the practical content is that a means test of the Australian kind leaves the saving policy monotone and puts a discontinuity in the consumption function at the free area. Solution methods that rely on monotonicity of the policy in assets are safe; methods that rely on a continuous or monotone consumption function, or on concavity of the value function, are not. That is close to the practice the applied literature has already settled on, which abandons Euler equation methods and searches a fine grid (Wellschmied, 2021; Garstenauer and Siassi, 2024), and the results here say which half of that caution is needed. For theory the content is that the two state spaces are not interchangeable once the pension depends on assets, and that monotone comparative statics survives non-concavity far better than it is usually given credit for.

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A Index of formal results

All in the namespaces `LeanEconomics.OLG` and `LeanEconomics.IncomeFluctuation`, files `LeanEconomics/OLG/MeansTest.lean` and `Retirement.lean`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	File
Retired household, lump-sum pension	<code>withPension</code>	Retirement
Human wealth is shock-independent	<code>stageHumanWealth_</code> <code>withPension_</code> <code>const</code>	Retirement
Perpetuity bound $H_k \leq p/r$	<code>stageHumanWealth_withPension_le</code>	Retirement
Cash on hand rises with assets	<code>withPension_resources_strictMono</code>	Retirement
Pension schedules	<code>lumpSum</code> , <code>tapered</code> , <code>cliff</code>	MeansTest
Cash on hand	<code>cash</code>	MeansTest
One period of the problem	<code>isOptimalSaving</code>	MeansTest
Theorem 3.1	<code>isOptimalSaving_mono</code>	MeansTest
Monotone in assets when cash is	<code>isOptimalSaving_mono_assets</code>	MeansTest
Theorem 3.2, the two optima	<code>isOptimalSaving_</code> <code>six</code> , <code>isOptimalSaving_eight</code>	MeansTest
Theorem 3.2	<code>exists_consumption_decrease</code>	MeansTest
Consumption is cash less saving	<code>consumptionFnOf_eq_sub</code>	MeansTest
Lump sum keeps cash rising	<code>cash_strictMono_lumpSum</code>	MeansTest
Positive part is one-Lipschitz	<code>max_zero_sub_le</code>	MeansTest
Theorem 4.2, gentle taper	<code>cash_strictMono_tapered</code>	MeansTest
Theorem 4.2, cliff reverses cash	<code>cash_lt_cash_of_cliff</code>	MeansTest
Theorem 4.2, the two optima	<code>isOptimalSaving_</code> <code>two</code> , <code>isOptimalSaving_threeHalves</code>	MeansTest
Theorem 4.2	<code>exists_saving_decrease_in_assets</code>	MeansTest
Theorem 4.1	<code>policyOf_mono_of_lt</code>	MeansTest